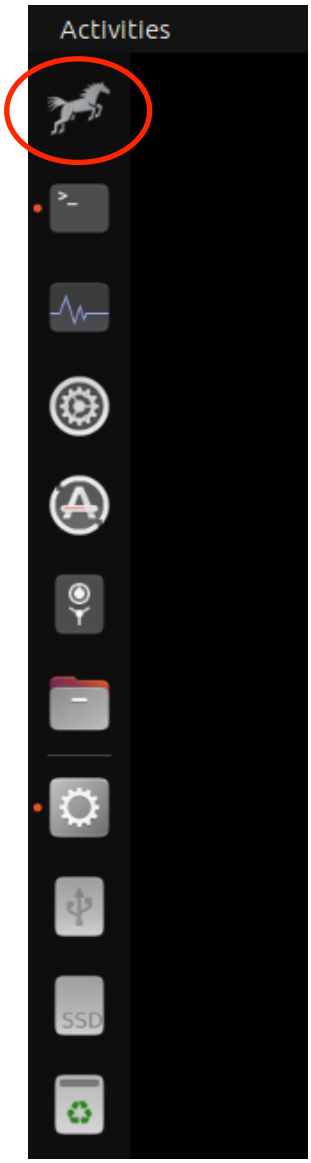
DHN Acq Manual

DHN Acq

Figure 1:



DHN Acq is software to acquire data from acquisition hardware directly into the MED format. Currently DHN Acq is only configured to acquire from Neuralynx systems, but this is not it's ultimate intended scope.

DHN Acq is a graphical user interface (GUI) acquisition, with no command line usage required. It is launched by clicking the DHN Dock icon (figure 1).

DHN Acq relies on two text files for configuration:

CHANNEL SPECIFICATION (CS) FILE

This file, which enumerates channel names, block sizes, decimation frequencies, compression parameters, & alerts for the channels to be recorded. This file will be described in more detail below. If no channel specification file is provided, DHN_Acq will use the default channel specification file named "DHN_Acq_cs.csv" in the current directory. If the default file does not exist, DHN_Acq will use default channel names, and record all channels at the incoming sampling frequency with no decimation.

The cs file is formatted as csv (comma separated values) and includes a header row for easy creation or editing in a spreadsheet program. Depending on typical usage, a new cs file may need to be created for each recording, but if not, it can certainly be reused.

RUNTIME CONFIGURATION (RC) FILE

This file specifies numerous details about how recording should be performed. This file will be described in more detail below. If no runtime configuration file is provided, DHN_Acq will use the default runtime configuration file named "DHN_Acq_rc.txt" in the current directory. If the default file does not exist, DHN_Acq will exit.

The runtime configuration (rc) file can be specified once for standard recording requirements, or a set of rc files can be kept for a various recording requirements which can then be specified at recording startup.

Start Up

Figure 2:

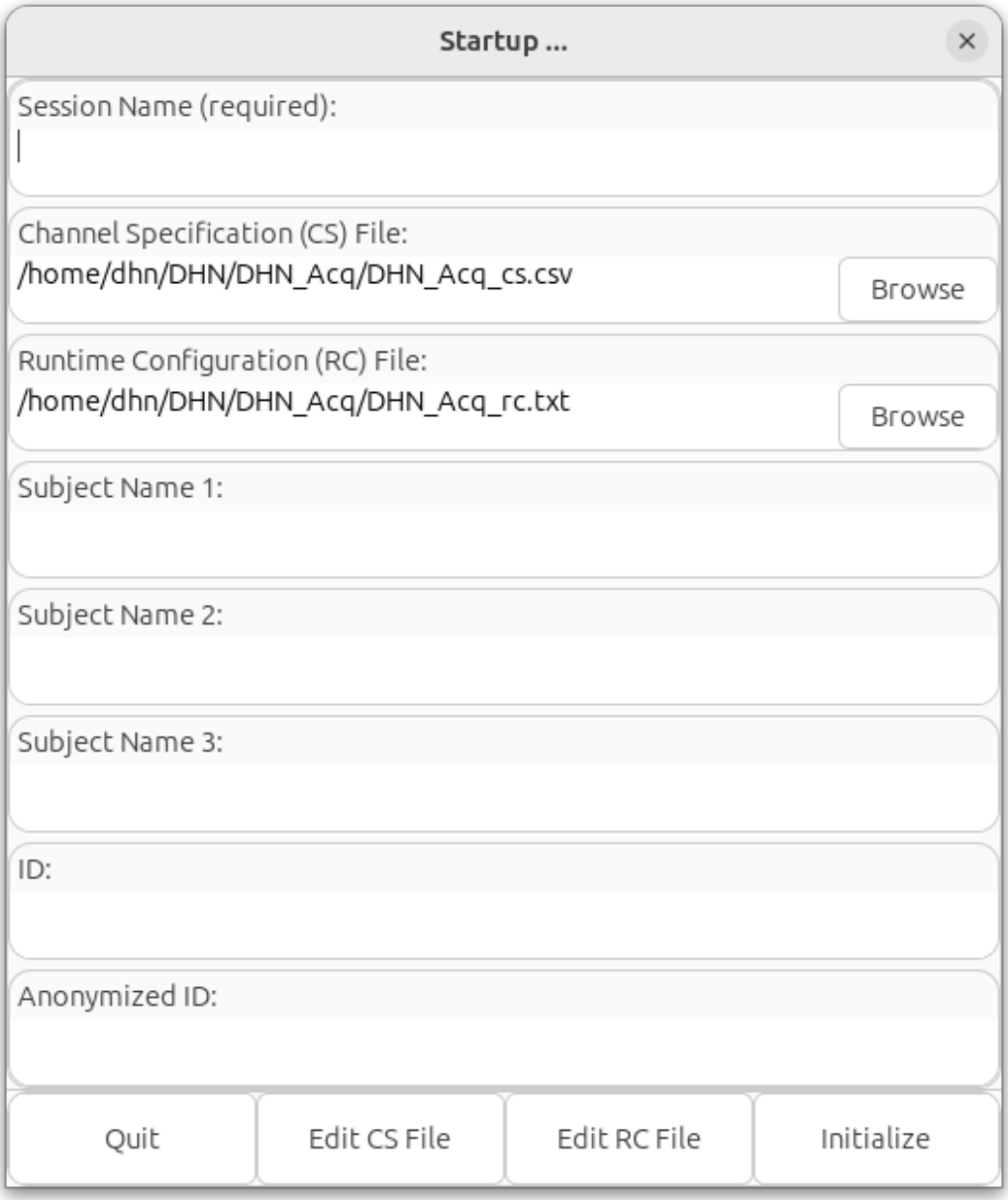
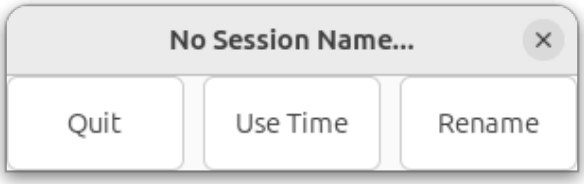


Figure 3:



At startup DHN Acq presents the dialog shown in figure 2 with the following fields:

SESSION NAME
This is required, no default value is assumed. This is the name of the MED session to be recorded (without the ".medd" extension). If the "Initialize" is pushed without a session name, the dialog in figure 3 will be displayed.

CHANNEL SPECIFICATION FILE
This is the name of the channel specification (cs) file, to be use for this recording. The default CS file is set by the dialog, but can be changed either by typing in the field or clicking on the browse button.

RUNTIME CONFIGURATION FILE
This is the name of the runtime configuration (rc) file, to be use for this recording. The default RC file is set by the dialog, but can be changed either by typing in the field or clicking on the browse button.

NAME 1, NAME 2, NAME 3
These are subject names. They are typically used as first, middle, and last names, if the subject is a person. These names are stored in section 3 of the MED metadata files, along with other potentially subject identifying data, and can be encrypted separately from technical recording details. If not specified, these fields are simply empty.

ID
These is a subject ID. If the subject is a human patient, it is typically used to store a medical record number, but any alphanumeric string is acceptable. It is stored in section 3 of the MED metadata files, along with other potentially subject identifying data, and can be encrypted separately from technical recording details. If not specified, this field is simply empty.

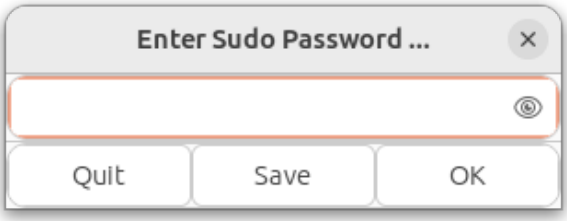
ANONYMIZED ID
This is an anonymized subject identifying field. Anonymization is external, not a function of MED. An anonymized name or number are typical contents, as in a research protocol. It is stored as a string. This is stored in the universal header of every MED file, and *cannot be encrypted*. If not specified, this field is simply empty.

The startup dialog allows creation of a CS file via the "Create CS File" button, and editing of the specified RC file via the "Edit RC" button. These are discussed below.

The "Initialize" button begins acquisition.

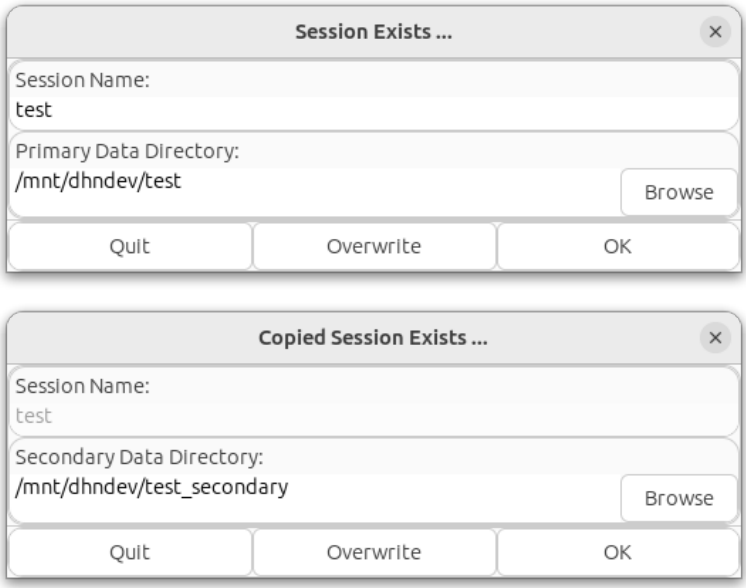
Sudo Password

Figure 4:



On first, and possibly every, launch, a dialog requesting the sudo password is presented (figure 4). DHN Acq requires superuser access to alter various system parameters that optimize recording performance. If, after entering this password, you select "Save", the password will be saved in encrypted form in a hidden file for future launches, and this dialog will not be presented again. Selecting "OK" will use the password for this session, but will not store it.

Figure 5:

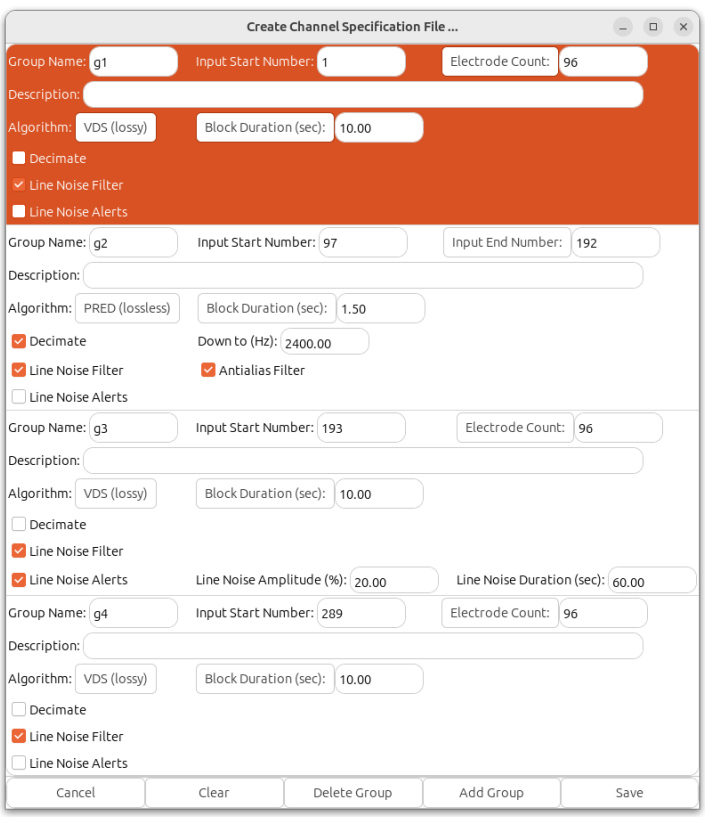


If the session exists on the primary or secondary storage directories the dialogs in figure 5 will be displayed.

You cannot specify a different name for the copied session.

Edit CS File

Figure 6:



The “Edit CS File” button brings up the dialog shown in figure 6. The CS file displayed is the one listed in the startup dialog.

Groups are displayed as rows in the table. The “Add Group” button adds a new group. The “Delete Group” button delete the currently selected group. Use the “Clear” button to create a new CS file.

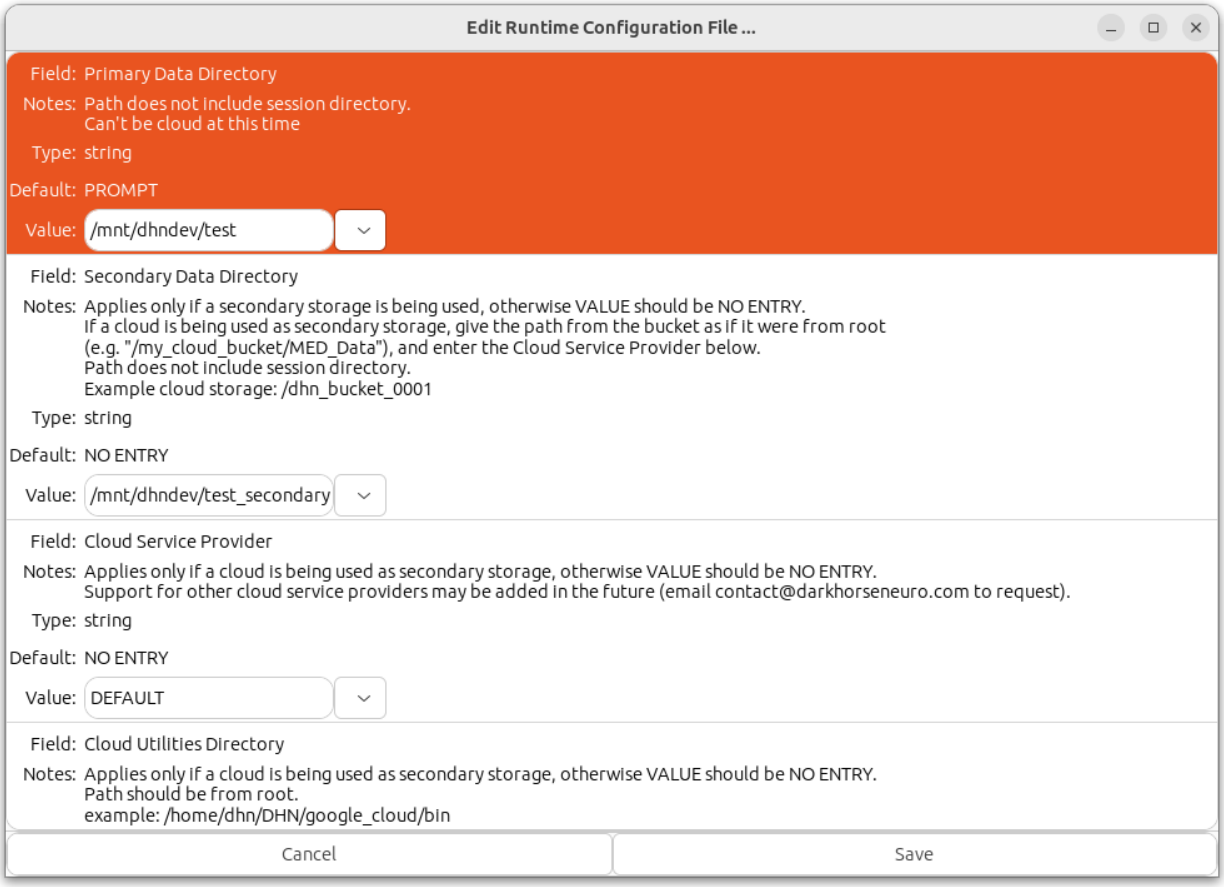
The channel names will be generated from the group name and the electrode count. For example the first channel of prefix “group1” will be named “group1_0001”.

All members of a group will have the same parameters. Member-specific parameters can be achieved by editing the resultant CS file with a text editor or spreadsheet program.

Groups options change dynamically to those relevant to the current configuration.

Edit RC File

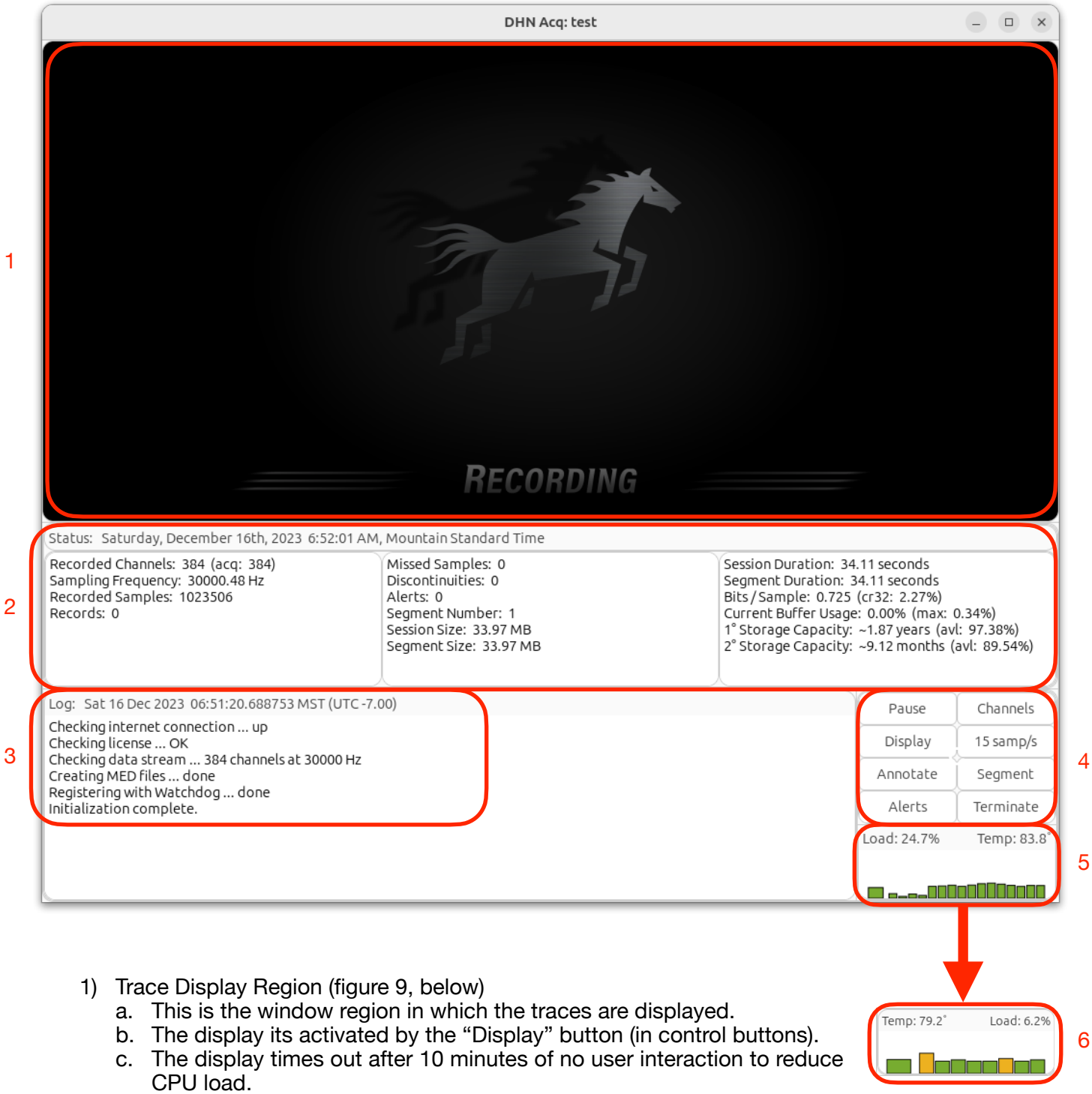
Figure 7:



The entries in the specified RC file will be displayed as rows in a table that can be edited and saved. For those familiar with the RC format, the “PROMPT” RC value option is not handled in this version of DHN_Acq.

Main Window

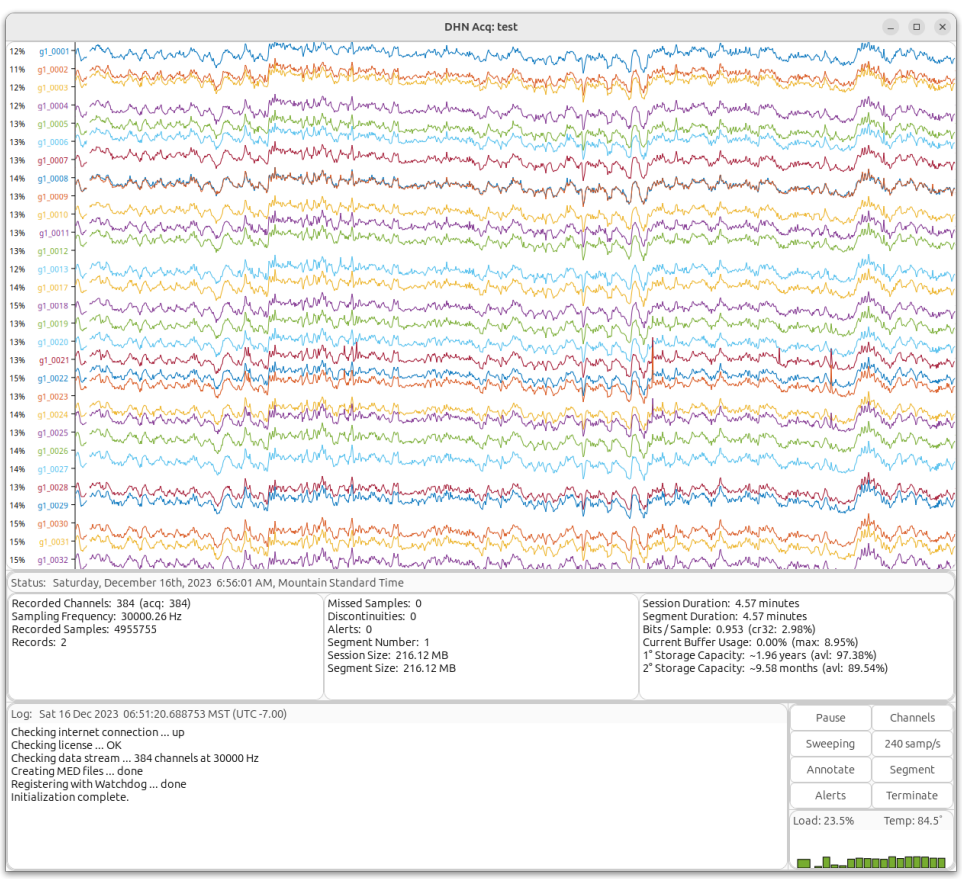
Figure 8:



1) Trace Display Region (figure 9, below)
   a. This is the window region in which the traces are displayed.
   b. The display its activated by the “Display” button (in control buttons).
   c. The display times out after 10 minutes of no user interaction to reduce CPU load.
   d. When not displaying traces, the image displays whether the system is acquiring or recording.

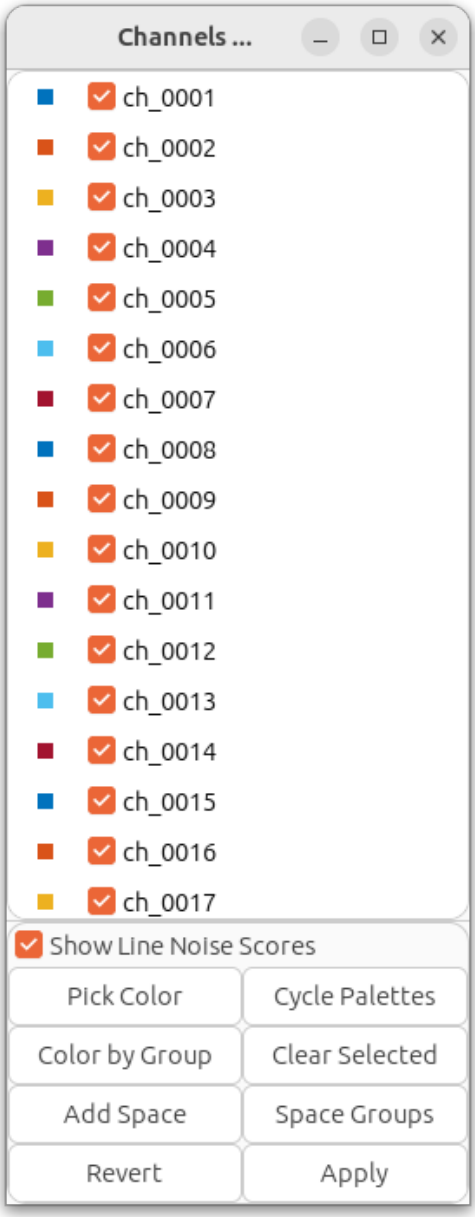
2) Recording Status
   a. Displays information about the current recording.
   b. The data is updated every 10 seconds.
3) Recording Log
   a. Displays real time messages from the acquisition system.
4) Control Buttons
   a. Record / Pause: Starts & pauses recording.
   b. Channels: Displays a dialog to select channel display parameters (described below).
   c. Display / Sweep / Scroll: Toggle between no display, sweeping, & scrolling. Left click cycles forward, right click cycles backward.
   d. Display Rate: Toggles the display rate. Left click cycles forward, right click cycles backward.
   e. Annotate: Brings up a dialog to add an annotation to the session.
   f. Segment: Brings up a dialog to create a new session segment.
   g. Alerts: Brings up a dialog to start & stop alerts and to change the alert email or text number.
   h. Terminate: Brings up a dialog to stop acquisition.
5) CPU monitor (load mode): Displays current CPU loads. The first, wider bin and text is the average, followed by the logical cores. The average core temperature is also displayed.
6) CPU monitor (temperature mode): Displays current CPU temperatures. The first, wider bin and text is the average, followed by the physical cores. The average core load is also displayed.
7) Toggle between CPU Monitor modes by clicking on the plot.

Figure 9:



Channel Display Options

Figure 10:



Pushing the “Channels” button in the main display brings up the dialog shown in figure 10.

• Channels are selected by clicking on their color swatches; selected channels have a grayed box around their swatches.

• Shift-select will select all channels between the clicked and last selected channels. This works on the check marks as well as the swatches.

• The Control key will select all channels. Deselect all select channels with the dialog’s “Clear Selected” button.

• A check beside the channel name indicates that it will be displayed. This also applies to spaces if present.

• The “Show Line Noise Scores” will show the line noise percentage of the channels at the left of the trace window (figure 9). Values in red exceed the line noise amplitude threshold set for the channel.

• The “Pick Color” button will bring a color chooser dialog and the selected color will be applied to all selected channels.

• The “Cycle Palettes” button will rotate the color palettes over all channels in one of four modes. Repeated clicking cycles to the next mode.

• The “Color by Group” button will assign color to the channels by group.

• The “Clear Selected” button will deselect all selected channels.

• The “Add Space” button will add a space after all selected channels.

• The “Space Groups” button will add a space after all selected groups (or all groups if no channels are selected).

• The “Revert” button will set all parameters to the state they were in when the dialog was launched.

• The “Apply” button will apply the changes to the current display and save the parameters as the current set.

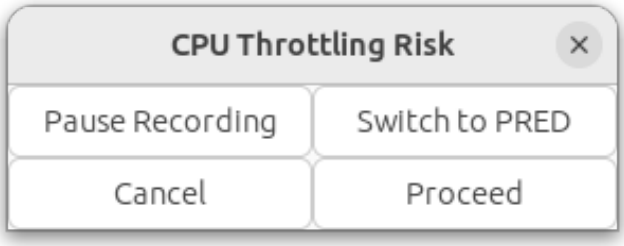
Trace Display Options

In the main window (sweep or scroll display mode):

- Up-arrow increases trace amplitudes, maintaining their centers. Shift-up-arrow does the same at a faster rate.
- Down-arrow decreases trace amplitudes, maintaining their centers. Shift-down-arrow does the same at a faster rate.
- Control-up-arrow increases trace spacing. This may push some traces off the display, but the display can then be scrolled by a scroll bar on the right. Shift-control-up-arrow does the same, but at a faster rate.
- Control-down-arrow decreases trace spacing, Shift-control-down-arrow does the same, but at a faster rate.

CPU Throttling

Figure 11:



CPU throttling can happen under some circumstances, & if it does, the degree to which the CPUs are slowed can result in buffer backlog and eventual overflow.

The system can tolerate some backlog, and the buffer size can be increased in the RC file. However, recovery to normal speeds usually takes too long for typical buffer lengths.

Predicting throttling with precision is difficult as it depends on many factors. The most common of these are:

1. Low number of CPU cores
2. High number of displayed channels
3. Large channels spacing (large scrollable channel area)
4. Scrolling (vs sweeping)
5. High number of channels compressed with VDS
6. Line noise filtering
7. High sampling frequencies
8. No decimated channels
9. High ambient temperature

However, two factors are absolutely required for it to happen:

1. Display is on
2. At least some channels are being recorded with VDS, and recording

If the CPU monitor detects throttling starting it will automatically disable the display, which is usually adequate to prevent it. A dialog is shown to alert you that this was done purposefully.

If you choose to display, or record under these circumstances, you will see the dialog shown in figure 11. You have five choices:

1. Pause Recording: The dialog was triggered by trying to display, so pause recording & continue with display.
2. Pause Display: The dialog was triggered by trying to record, so pause display & continue with recording.
3. Switch to PRED: All VDS channels will be switched to PRED, temporarily, until the throttling risk is gone (e.g. display turned off). After that they will automatically switch back to VDS. You will not lose any data, but compression ratios will be impacted for the duration of the switch. You do not have to switch back manually.
4. Cancel: Don't start display or recording, whichever you were trying to do.
5. Proceed: This is not absolutely a bad choice. Throttling is rare, and experience may well have taught you that your current configuration will not cause it on your machine.

Runtime Configuration (RC) File

RC File Structure:

Any line not beginning with "%%" is considered a comment and ignored by the function.

An Entry consists of "FIELD", "NOTES", "TYPE", "OPTIONS", "DEFAULT", and "VALUE" labels.
"TYPE", "OPTIONS", "DEFAULT", and "VALUE" labels must exist for each entry.

"NOTES" labels are optional. Any number of "NOTES" lines may appear in an entry.
The other labels are restricted to one line.

This order is required in an entry:
%% FIELD:
%% NOTES: (optional)
%% TYPE:
%% OPTIONS:
%% DEFAULT:
%% VALUE:

Entry "TYPE"s are one of: string, float, integer, ternary.

The string "NO ENTRY" can be entered as a DEFAULT or VALUE where appropriate.
If "NO ENTRY" is entered as the value:
a) string types will return zero-length strings
b) float types will return 0.0
c) integer types will return 0
d) ternary types will return "UNKNOWN"

"DEFAULT" values are used when there is no text following in the value label or the word "DEFAULT" is specified as the value.

"PROMPT" will ask the user for input and present the OPTIONS and DEFAULT.
The user can also enter the strings "DEFAULT", "NO ENTRY", or any of the defined OPTION strings, as well as free text if appropriate.

"PROMPT" and "NO ENTRY" are not considered OPTIONs, but may serve as values or defaults.

The "OPTIONS" label may be modified to "OPTIONS ONLY", in which case, only the listed options or "NO ENTRY", "DEFAULT", or "PROMPT" are permitted as values.

DHN_Acq RC File Contents:

```
%% FIELD: Primary Data Directory
%% NOTES: Path does not include session directory.
%% NOTES: Can't be cloud at this time
%% TYPE: string
%% OPTIONS:
%% DEFAULT: PROMPT
%% VALUE: /mnt/dhndev/test

%% FIELD: Secondary Data Directory
%% NOTES: Applies only if a secondary storage is being used, otherwise VALUE should be NO ENTRY.
%% NOTES: If a cloud is being used as secondary storage, give the path from the bucket as if it
were from root
%% NOTES: (e.g. "/my_cloud_bucket/MED_Data"), and enter the Cloud Service Provider below.
%% NOTES: Path does not include session directory.
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: /mnt/dhndev/test_secondary

%% FIELD: Cloud Service Provider
%% NOTES: Applies only if a cloud is being used as secondary storage, otherwise VALUE should be
NO ENTRY.
%% NOTES: Support for other cloud service providers may be added in the future (email
contact@darkhorseneuro.com to request).
%% TYPE: string
%% OPTIONS: GOOGLE, AMAZON
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Cloud Utilities Directory
%% NOTES: Applies only if a cloud is being used as secondary storage, otherwise VALUE should be
NO ENTRY.
%% NOTES: Path should be from root.
%% NOTES: example: /home/dhn/DHN/google_cloud/bin
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Channel Specification File
%% NOTES: If not specified, all channels will be recorded at the incoming sampling frequency,
with default filenames.
%% NOTES: Build file with “create_channel_specs” or in spreadsheet or text editor program.
%% NOTES: Default output file name from “create_channel_specs” is “DHN_Acq_cs.csv”
%% TYPE: string
%% OPTIONS:
%% DEFAULT: DHN_Acq_cs.csv
%% VALUE: DEFAULT

%% FIELD: Allow Segmenting
%% NOTES: Segmenting requires more memory
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT
```

%% FIELD: Auto-Segment Data
%% NOTES:
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Auto-Segment Interval
%% NOTES: segment duration of format hh:mm
%% NOTES: interval timing begins at recording start
%% NOTES: this value supersedes "Auto-Segment Time" (below), if set
%% NOTES: set to NO ENTRY here to specify an auto-segment time
%% NOTES: auto-segmenting does not prevent manual segmenting, however a manual segment resets this timer to zero
%% NOTES: units in hours
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Auto-Segment Time
%% NOTES: 24 hour time of format hh:mm in local time
%% NOTES: this option is superseded by "Auto-Segment Interval" (above), if set
%% NOTES: auto-segmenting does not preclude manual segmenting; auto-segmenting will continue at the specified time
%% NOTES: "SEGMENT AT MIDNIGHT" is equivalent to setting this parameter to 00:00
%% NOTES: this option creates 24 hour segments (except the first one, and manual segments)
%% TYPE: string
%% OPTIONS: SEGMENT AT MIDNIGHT
%% DEFAULT: SEGMENT AT MIDNIGHT
%% VALUE: DEFAULT

%% FIELD: Auto-Copy Segments
%% NOTES: Segments can be auto-copied to secondary file system after closing them
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Auto-Delete Copied Segments
%% NOTES: If segments are copied they can be deleted from the main file system when the copy is complete
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Packet Buffer Seconds
%% NOTES: This applies only to acquisition with segmenting allowed.
%% NOTES: If segmenting is done, the buffer size needed is dependent on the speed of the media.
%% NOTES: Points of reference (with 512 channels): slow external drive can require up to 90 sec, fast internal SSD requires about 3 sec.
%% NOTES: Just good practice to consume less OS resources, within reason.
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 15.0
%% VALUE: DEFAULT

%% FIELD: Session Exists Behavior
%% NOTES:
%% TYPE: string
%% OPTIONS ONLY: PROMPT IF EXISTS, OVERWRITE, QUIT
%% DEFAULT: PROMPT IF EXISTS
%% VALUE: DEFAULT

%% FIELD: Include Segmented Session Records
%% NOTES: Divide appropriate session-level records into separate files for each segment
%% NOTES: SyLg (System Log) and Sgmt (Segment) records will go into the standard session records file
%% NOTES: Note (Annotations) and NlxP (Neuralynx Port) records will be go into their respective segmented record files
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Robust Mode
%% NOTES: Update file headers with every write, files always intact in case of catastrophic failure
%% NOTES: Significantly more write overhead and disk wear
%% NOTES: THIS OPTION IS NOT FULLY IMPLEMENTED AT THIS TIME
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Block Samples
%% NOTES: 25000 to 75000 samples per block is a good range for most purposes.
%% NOTES: Another common choice is to select USE BLOCK DURATION here,
%% NOTES: and set block duration to a fixed number such as 1.0 second.
%% NOTES: If a numerical value is set here, it supersedes any value entered into
%% NOTES: the "Block Duration" field.
%% NOTES: If VDS compression is selected, longer blocks are recommended (e.g. select USE BLOCK DURATION here,
%% NOTES: and set "Block Duration" to >= 10.0 seconds below)
%% TYPE: integer
%% OPTIONS: USE BLOCK DURATION
%% DEFAULT: 50000
%% VALUE: DEFAULT

%% FIELD: Block Duration
%% NOTES: units of seconds
%% NOTES: If a value is set in the "Block Samples" field, it supersedes any value entered here.
%% TYPE: float
%% OPTIONS: USE BLOCK SAMPLES
%% DEFAULT: USE BLOCK SAMPLES
%% VALUE: DEFAULT

%% FIELD: Session Description
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

```
%% FIELD: Equipment Description
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: Neuralynx Atlas / Pegasus acquisition system

%% FIELD: Reference Description
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Antialias Filter
%% NOTES: roll off starts at 4 samples per cycle
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Line Noise Filter
%% NOTES: reduces line noise without attenuating physiologic signal in the same band
%% NOTES: this option performs better with longer blocks - e.g. 10 seconds
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Line Noise Alerts
%% NOTES: send alerts for sustained high line noise
%% NOTES: threshold for alert set with "Line Noise Alert Threshold" (below)
%% NOTES: THIS OPTION IS NOT CURRENTLY ENABLED
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Line Noise Alert Amplitude Threshold
%% NOTES: percentage of signal amplitude accounted for by line noise
%% NOTES: that should trigger an alert (if "Line Noise Alerts" is set - above)
%% NOTES: THIS OPTION IS NOT CURRENTLY ENABLED
%% NOTES: range 0 to 100%
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 20.0
%% VALUE: DEFAULT

%% FIELD: Line Noise Alert Duration Threshold
%% NOTES: time the of signal amplitude accounted for by line noise should be above the amplitude
%% NOTES: threshold before an alert should be triggered
%% NOTES: THIS OPTION IS NOT CURRENTLY ENABLED
%% NOTES: units in seconds
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 60.0
%% VALUE: DEFAULT
```

```
%% FIELD: Low Frequency Filter Setting
%% NOTES: units in Hz
%% TYPE: float
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: 0.0

%% FIELD: High Frequency Filter Setting
%% NOTES: units in Hz
%% NOTES: 9 kHz is analog antialiasing filter cutoff in Nlx hardware
%% NOTES: If VDS compression is selected (below) with LFP filtering enabled,
%% NOTES: the lowpass cutoff can be set here with the USE HIGH FREQUENCY FILTER SETTING
%% NOTES: option, in the VDS settings (below)
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 9000.0
%% VALUE: DEFAULT

%% FIELD: Notch Filter Frequency Setting
%% NOTES: units in Hz
%% TYPE: float
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: AC Line Frequency
%% NOTES: Europe / Africa / Asia / Australia mostly 50 Hz Americas mostly 60 Hz
%% NOTES: units in Hz
%% TYPE: float
%% OPTIONS: 50.0, 60.0
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Amplitude Units Conversion Factor
%% NOTES:
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 1.0
%% VALUE: DEFAULT

%% FIELD: Amplitude Units Description
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: microvolts
%% VALUE: DEFAULT

%% FIELD: Time Base Units Conversion Factor
%% NOTES:
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 1.0
%% VALUE: DEFAULT
```

%% FIELD: Time Base Units Description
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: microseconds
%% VALUE: DEFAULT

%% FIELD: Apply Recording Time Offset
%% NOTES:
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Standard Timezone Acronym
%% NOTES: e.g "MST" for "Mountain Standard Time"
%% NOTES: not shorter synonym e.g. "MT" for "Mountain Time"
%% NOTES: not the Daylight Saving / Summer Time version e.g. "MDT" for "Mountain Daylight Time"
%% NOTES: this field is used define the correct timezone for territories containing multiple timezones
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: MST

%% FIELD: Recording Country
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: United States

%% FIELD: Recording Territory
%% NOTES: state, province, territory, etc.
%% NOTES: In the US (only), the territory is required to accurately define Daylight Saving Time behavior as some states do not observe it.
%% NOTES: In the US (only), enter the two-letter state abbreviation or full name of the state.
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: Montana

%% FIELD: Recording Locality
%% NOTES: city, township, village, etc.
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: Bozeman

%% FIELD: Recording Institution
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: Dark Horse Neuro

%% Encryption NOTES:
%% LEVEL 0 LEVEL 1 LEVEL 2
%% Level 0 encryption is equivalent to no encryption
%% Level 1 encryption requires specification of a LEVEL 1 password
%% Level 2 encryption requires specification of a LEVEL 2 password and a LEVEL 1 password (even if LEVEL 1 encryption is not used anywhere in the file).
%% Level 2 passwords provide access to both level 1 & 2 encrypted fields. (Level 1 passwords do not need to be explicitly entered.)
%% The software will determine whether the entered password is Level 1 or Level 2; this does not need to be specified.

%% FIELD: Metadata Section 2 Encryption Level
%% NOTES: technical data about the recording
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 1
%% VALUE: DEFAULT

%% FIELD: Metadata Section 3 Encryption Level
%% NOTES: data source identifying data
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 2
%% VALUE: DEFAULT

%% FIELD: Time Series Data Encryption Level
%% NOTES:
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 0
%% VALUE: DEFAULT

%% FIELD: Segment Record Encryption Level
%% NOTES: Typically the same as Metadata Section 2 Encryption Level
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 1
%% VALUE: DEFAULT

%% FIELD: Neuralynx Port Record Encryption Level
%% NOTES: Neuralynx parallel port events
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 0
%% VALUE: DEFAULT

%% FIELD: Annotation Record Encryption Level
%% NOTES: User entered notes during acquisition
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 2
%% VALUE: DEFAULT

%% FIELD: System Log Record Encryption Level
%% NOTES: System entered data during acquisition
%% TYPE: string
%% OPTIONS ONLY: LEVEL 0, LEVEL 1, LEVEL 2
%% DEFAULT: LEVEL 0
%% VALUE: DEFAULT

%% FIELD: Level 1 Password
%% NOTES: NO ENTRY is equivalent to LEVEL 0 encryption (no encryption)
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: L1_password

%% FIELD: Level 1 Password Hint
%% NOTES: presented to user if incorrect password is entered
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: the level 1 password is 'L1_password'

%% FIELD: Level 2 Password
%% NOTES: NO ENTRY does not prevent LEVEL 1 encryption
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: L2_password

%% FIELD: Level 2 Password Hint
%% NOTES: presented to user if incorrect password is entered
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: the level 2 password is 'L2_password'

%% FIELD: Level 3 Password
%% NOTES: Specification of a Level 3 Password is a failsafe mechanism for retrieval of level 1 & level 2 passwords
%% NOTES: The "USE DHN L3 PW" option will encode using a proprietary Dark Horse Neuro password.
%% NOTES: DHN can retrieve lost passwords if this is option is selected.
%% TYPE: string
%% OPTIONS: USE DHN L3 PW
%% DEFAULT: NO ENTRY
%% VALUE: USE DHN L3 PW

%% FIELD: Include Session-level Segment Records
%% NOTES: tiny footprint, significantly improves search performance
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Include Channel-level Segment Records
%% NOTES: tiny footprint, significantly improves search performance
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

```
%% FIELD: Include Neuralynx Port Records
%% NOTES:
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Neuralynx Port Trigger Mode
%% NOTES:
%% TYPE: string
%% OPTIONS ONLY: NO TRIGGER, ANY CHANGE, HIGH BIT SET
%% DEFAULT: HIGH BIT SET
%% VALUE: DEFAULT

%% FIELD: Neuralynx Port Zero is Reset
%% NOTES: In NO TRIGGER and ANY CHANGE trigger modes, subport values of zero do not generate records.
%% NOTES: This allows NO TRIGGER mode to store values only when there the subport is non-zero.
%% NOTES: This allows ANY CHANGE mode to store onset value / time pairs only, when offset times are not required, reducing the number of records.
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Number of Subports in Neuralynx Port
%% NOTES: no subports (full 32 bits) == 1 subport in this context
%% TYPE: integer
%% OPTIONS ONLY: 1, 2, 4
%% DEFAULT: 2
%% VALUE: 1

%% FIELD: Remote Notification Email Address
%% NOTES:
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Remote Notification Email Addressee
%% NOTES: recipient's name
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT

%% FIELD: Remote Notification Text Number
%% NOTES: hyphens, spaces, parentheses, and commas will be stripped out
%% NOTES: plus signs will be left in
%% TYPE: string
%% OPTIONS:
%% DEFAULT: NO ENTRY
%% VALUE: DEFAULT
```

%% FIELD: Remote Notification for Alerts
%% NOTES:
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Remote Alert Notification Medium
%% NOTES:
%% TYPE: string
%% OPTIONS ONLY: TEXT, EMAIL, TEXT AND EMAIL
%% DEFAULT: TEXT AND EMAIL
%% VALUE: DEFAULT

%% FIELD: Use Watchdog Service
%% NOTES: Use the DHN watchdog server to alert if the acquisition software terminates unexpectedly (e.g. power outage, process killed).
%% NOTES: The alert will be communicated to the alert recipient via the alert notification media, specified above or in the software.
%% NOTES: At least one notification medium must exist to use the service.
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: Remote Notification for Updates
%% NOTES:
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Remote Update Notification Medium
%% NOTES:
%% TYPE: string
%% OPTIONS ONLY: TEXT, EMAIL, TEXT AND EMAIL
%% DEFAULT: EMAIL
%% VALUE: DEFAULT

%% FIELD: Remote Update Notification Frequency
%% NOTES: units in hours
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 1.0
%% VALUE: DEFAULT

%% FIELD: Neuralynx PC MAC Address
%% NOTES: Required field.
%% NOTES: On the Neuralynx acquisition PC, in a terminal (Command Prompt) type "ipconfig /all <enter>"
%% NOTES: and get the MAC address associated with the "192.168.3.100" IP address.
%% NOTES: ipconfig formats MAC address with hyphens, e.g. 00-00-00-00-00-00, substitute with colons here
%% NOTES: the MAC address is not necessary in local broadcast mode
%% TYPE: string
%% OPTIONS:
%% DEFAULT:
%% VALUE: 00:00:00:00:00:00

%% FIELD: Receive As Broadcast
%% NOTES: connection from Neuralynx, or forwarder
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: YES

%% FIELD: Receiving Local EN Interface Name
%% NOTES: connection from Neuralynx, or forwarder
%% TYPE: string
%% OPTIONS:
%% DEFAULT: eno1
%% VALUE: DEFAULT

%% FIELD: Receiving Local IP Address
%% NOTES: local IP connected to Neuralynx acquisition box, or forwarding server
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 192.168.3.100
%% VALUE: DEFAULT

%% FIELD: Receiving Server IP Address
%% NOTES: IP address of Neuralynx acquisition box (192.168.3.10), or forwarding server
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 192.168.3.10
%% VALUE: DEFAULT

%% FIELD: Receiving Port Number
%% NOTES: connection from Neuralynx, or forwarding server
%% NOTES: 26090 is Neuralynx acquisition box (hard coded)
%% TYPE: integer
%% OPTIONS:
%% DEFAULT: 26090
%% VALUE: DEFAULT

%% FIELD: Receiving Subnet Mask
%% NOTES: on connection to Neuralynx, or forwarding server
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 255.255.255.0
%% VALUE: DEFAULT

%% FIELD: Forward Neuralynx Packets
%% NOTES: DHN_Acq can forward Neuralynx packets from a second ethernet interface, as if they were coming directly from Neuralynx
%% NOTES: You cannot forward to the Neuralynx acquisition PC with this (Nlx control connection is also on that line)
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

```
%% FIELD: Forward As Broadcast
%% NOTES: applies to "Forward Neuralynx Packets" option
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

%% FIELD: Forwarding Local EN Interface Name
%% NOTES: applies to "Forward Neuralynx Packets" option
%% TYPE: string
%% OPTIONS:
%% DEFAULT: eno1
%% VALUE: DEFAULT

%% FIELD: Forwarding Local IP Address
%% NOTES: applies to "Forward Neuralynx Packets" option
%% NOTES: choose a different subnet from Neuralynx input (e.g. if Nlx == 192.168.3.10, use
192.168.1.10 for forwarding)
%% NOTES: and use that subnet to receive the forwarded data
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 192.168.1.10
%% VALUE: DEFAULT

%% FIELD: Forwarding Client IP Address
%% NOTES: applies to "Forward Neuralynx Packets" option
%% NOTES: not necessary for "Forwarding Broadcast Mode" below
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 192.168.1.100
%% VALUE: DEFAULT

%% FIELD: Forwarding Port Number
%% NOTES: Cannot be the same as receiving port
%% NOTES: 26090 is Neuralynx acquisition box (hard coded - DO NOT USE THIS NUMBER)
%% TYPE: integer
%% OPTIONS:
%% DEFAULT: 26091
%% VALUE: DEFAULT

%% FIELD: Forwarding Subnet Mask
%% NOTES: applies to "Forward Neuralynx Packets" option
%% NOTES: on connection to forwarding clients
%% TYPE: string
%% OPTIONS:
%% DEFAULT: 255.255.255.0
%% VALUE: DEFAULT

%% FIELD: Non-blocking Forwarding
%% NOTES: applies to "Forward Neuralynx Packets" option
%% NOTES: use when forwarding is stalling main acquisition (e.g. extreme channel counts)
%% NOTES: may result in some packet drop on forwarding socket
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: YES
```

%% FIELD: Compression Algorithm
%% NOTES: RED: Range Encoded Derivatives => lossless
%% NOTES: PRED: Predictive Range Encoded Derivatives => lossless (slower than RED, but better compression)
%% NOTES: MBE: Minimal Bit Encoding => lossless (generally used as fall through algorithm for degenerate data)
%% NOTES: VDS: Vectorized Data Stream => lossy (highest compression level; see other VDS options below)
%% TYPE: string
%% OPTIONS ONLY: RED, PRED, MBE, VDS
%% DEFAULT: PRED
%% VALUE: DEFAULT

%% FIELD: Compression Algorithm Fall Through
%% NOTES: Currently applies only to RED & PRED compression.
%% NOTES: If MBE would be smaller, block is compressed with MBE.
%% NOTES: Generally only occurs in blocks with degenerate data.
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: YES
%% VALUE: DEFAULT

%% FIELD: VDS Threshold
%% NOTES: Applies only to VDS compression.
%% NOTES: Range 0.0 to 10.0
%% NOTES: 0 is lossless (redirects to PRED), 10 is very lossy.
%% TYPE: float
%% OPTIONS:
%% DEFAULT: 5.0
%% VALUE: DEFAULT

%% FIELD: VDS LFP Filter Cutoff
%% NOTES: Applies only to VDS compression.
%% TYPE: float
%% OPTIONS: USE HIGH FREQUENCY FILTER SETTING, NO FILTER
%% DEFAULT: USE HIGH FREQUENCY FILTER SETTING
%% VALUE: 500.0

%% FIELD: VDS Scale by Baseline
%% NOTES: Applies only to VDS compression.
%% NOTES: Scale data by block baseline width. Increases compression by ~30%.
%% NOTES: Creates higher compression & decreased fidelity in proportion to the baseline width on a block by block basis.
%% TYPE: ternary
%% OPTIONS ONLY: YES, NO
%% DEFAULT: NO
%% VALUE: DEFAULT

Channel Specification (cs) File

CS File Structure:

The CS file is a plain text file, formatted as comma-separated values (csv). These files can be created or edited in any spreadsheet or word-processing program. The 16 columns are:

1. Channel Number
2. Channel Name
3. Decimation Frequency
4. Block Samples
5. Block Duration
6. Compression Algorithm
7. VDS Threshold
8. VDS LFP Cutoff
9. VDS Scale by Baseline
10. Antialias Filter
11. Line Noise Filter
12. Line Noise Alerts
13. Line Noise Alert Amplitude
14. Line Noise Alert Duration
15. Channel Description

The column names are included in the CS file as a header row.

Field Details:

1. Channel Number
   - This is the physical input plug number.
   - MED numbering for these starts at 1.
2. Channel Name
   - This is the base name, no extension. The software will add MED extensions.
3. Decimation Frequency
   - The sampling frequency to which the channel should be decimated.
   - 0 (or not entry) here will disable decimation on the channel.
   - If decimation is enabled, a low-pass antialiasing filter will be applied to the raw data with it's roll-off starting at 1/4 the decimation frequency (minimum 4 samples/cycle), prior to decimating.
4. Block Samples
   - The number of samples in a compressed block for the channel.
   - 0 (or no entry) here will use the value specified in the Block Duration field, if specified, otherwise it will use the value in the RC file.
5. Block Duration
   - The block duration (in microseconds) for the channel.
   - If the Block Samples field is specified, that value supersedes this one.
6. Compression Algorithm
   - RED, PRED, MBE, or VDS
7. VDS Threshold
   - If VDS compression is specified, this is the threshold parameter (0-10) for the algorithm.
8. VDS LFP Cutoff
   - If VDS compression is specified, this is the LFP cutoff parameter for the algorithm.
9. VDS Scale by Baseline

- If VDS compression is specified, this determines whether the algorithm will scale blocks by their baseline noise level
- Values: -1 = false; 0 = use RC value; 1 = true

10. Antialias Filter
    - Apply an antialias filter to the raw data prior to compression.
    - Values: -1 = false; 0 = use RC value; 1 = true
11. Line Noise Filter
    - Apply a line noise filter to the raw data prior to compression.
    - Values: -1 = false; 0 = use RC value; 1 = true
12. Line Noise Alerts
    - Alert for increased line noise on this channel, using amplitude & duration parameters specified below.
    - Values: -1 = false; 0 = use RC value; 1 = true
13. Line Noise Alert Amplitude
    - Alert when line noise exceeds this percentage of the overall channel amplitude.
    - Used in combination with Line Noise Alert Duration parameter.
14. Line Noise Alert Duration
    - Alert when excess line noise persists for more than this many seconds.
    - Used in combination with Line Noise Alert Amplitude parameter.
15. Channel Description
    - No entry is required.
    - All punctuation is allowed (including commas).
    - Newlines are not allowed.
    - Maximum of 1023 ascii or 255 utf-8 characters in length.

Vectorized Data Stream (VDS) Compression

VDS is a lossy compression scheme that converts time series data into a series of scalable anchors. This data representation can be interpolated at arbitrary resolution. VDS compression is computationally expensive, but decompression is very fast.

VDS has four main parameters:

1. Threshold
2. Local Field Potential Lowpass Filter Cutoff (LFP High FC)
3. Scale by Baseline
4. Block Samples

Threshold:
This parameter ranges from 0.0 to 10.0 and governs the lossiness of the compression, and is inversely related to the resultant compression ratio (CR):

CR = compressed_size / uncompressed_size (usually expressed as a percentage)

A threshold of 0.0 is lossless, and under the hood is redirected to PRED compression. A threshold of 10 is very lossy, and very close to the maximum compression that the algorithm can achieve. Fractional threshold within the range are permitted for finer control (e.g. thresh = 5.5).

Local Field Potential Lowpass Filter Cutoff (LFP High FC):
VDS can optionally perform the following:
1. Excise transients
2. Low pass filter the resultant trace with the user specified LFP high fc
3. Restore the transients to the filtered signal

The result of this process is to reduce the baseline variance between transients, and thus provide higher compression. LFP filtering is optional.

Scale by Baseline:
This is a boolean parameter that scales the data by a measure of the baseline amplitude for each MED block. The baseline width is a good surrogate for the margin of error in the amplitude of the signal at each point in time. Scaling by baseline reduces signal fidelity by this margin or error resulting in better compression. In practice the compression increases by about 30% with with only minor reductions in signal fidelity. The computational cost is negligible.

Block Samples:
This parameter should typically be much larger than for lossless compression because the number of stored samples in VDS is dramatically reduced compared to the lossless compression algorithms. Lower stored samples results in poorer statistical models, and a proportionately greater volume of model data. This number can be tailored to your needs, but 10 second VDS blocks are a good rule of thumb.

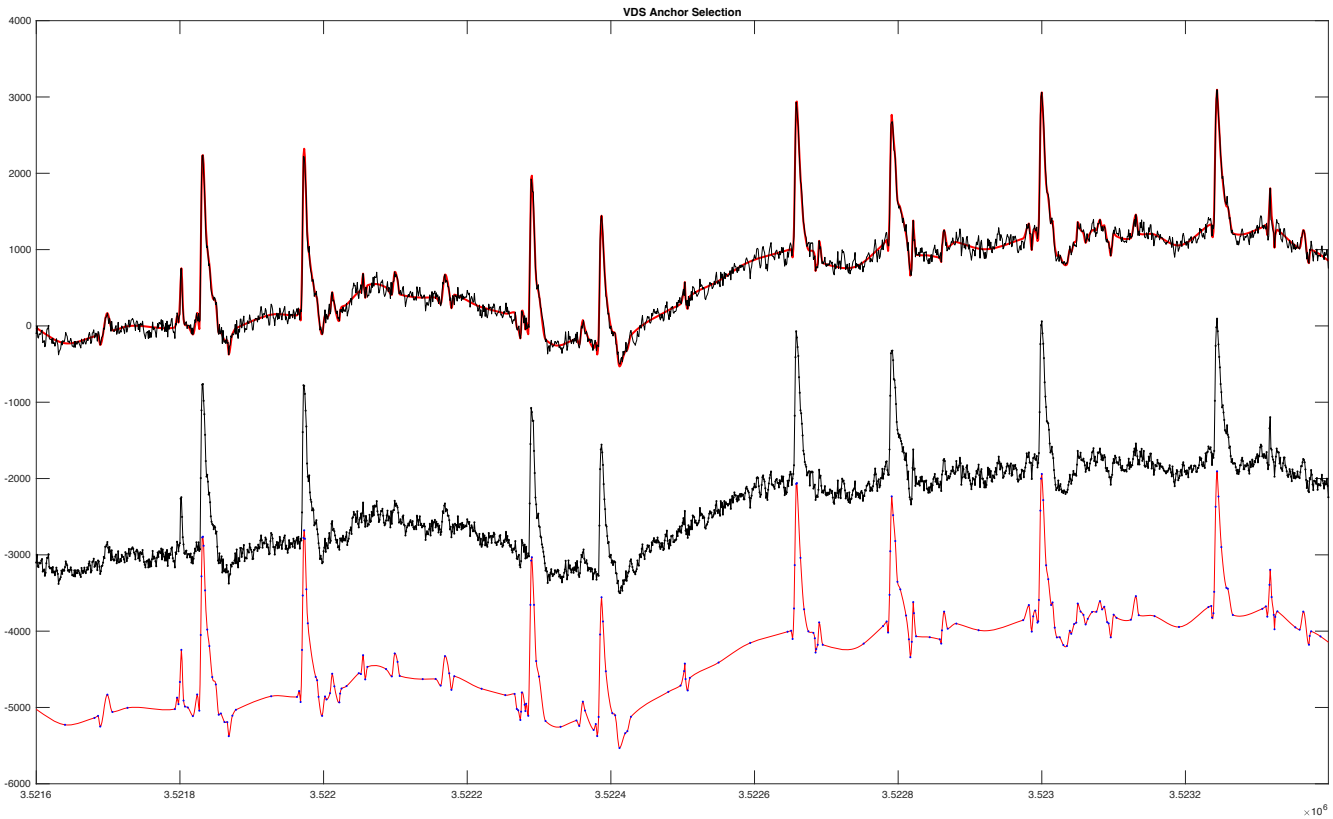
VDS Compression Table:

(compression ratios based on 16-bit input data)

Compression Ratio Comparison (384 channels, 30 KHz, 10 sec blocks, run on 8 Core 3 GHz Intel Xeon with 32 GB RAM & 2 TB internal SSD)

VDS Threshold (lossiness)	LFP Cutoff	Compression Speed (x faster than real time)	Compression Ratio	Compression Ratio (blocks scaled by baseline)
Lossless (-AA, -NFF)	n/a	49.18	37.01%	
Lossless (+AA, -NFF)	n/a	20.34	32.01%	
Lossless (-AA, +NFF)	n/a	18.97	23.65%	
Lossless (+AA, +NFF)	n/a	18.94	22.92%	
1	No Filter	1.40	27.26%	18.30%
2	No Filter	1.61	24.38%	17.04%
3	No Filter	1.80	21.60%	15.47%
4	No Filter	1.99	18.80%	13.68%
5	No Filter	2.24	16.00%	11.78%
6	No Filter	2.63	13.20%	9.94%
7	No Filter	2.63	10.40%	8.03%
8	No Filter	2.32	7.60%	6.01%
9	No Filter	1.98	4.80%	3.89%
10	No Filter	1.54	2.00%	1.65%
1	500	2.44	10.02%	7.31%
2	500	2.47	9.12%	6.70%
3	500	2.56	7.95%	5.89%
4	500	2.64	7.08%	5.28%
5	500	2.72	6.03%	4.53%
6	500	2.80	5.02%	3.80%
7	500	2.87	4.00%	3.04%
8	500	2.88	3.00%	2.30%
9	500	2.69	2.00%	1.54%
10	500	1.94	1.00%	0.81%
1	200	1.23	8.53%	6.11%
2	200	2.51	7.75%	5.59%
3	200	2.56	6.82%	4.96%
4	200	2.64	6.09%	4.46%
5	200	2.72	5.27%	3.89%
6	200	2.80	4.40%	3.28%
7	200	2.85	3.56%	2.67%
8	200	2.87	2.70%	2.04%
9	200	2.74	1.78%	1.35%
10	200	2.31	0.94%	0.76%

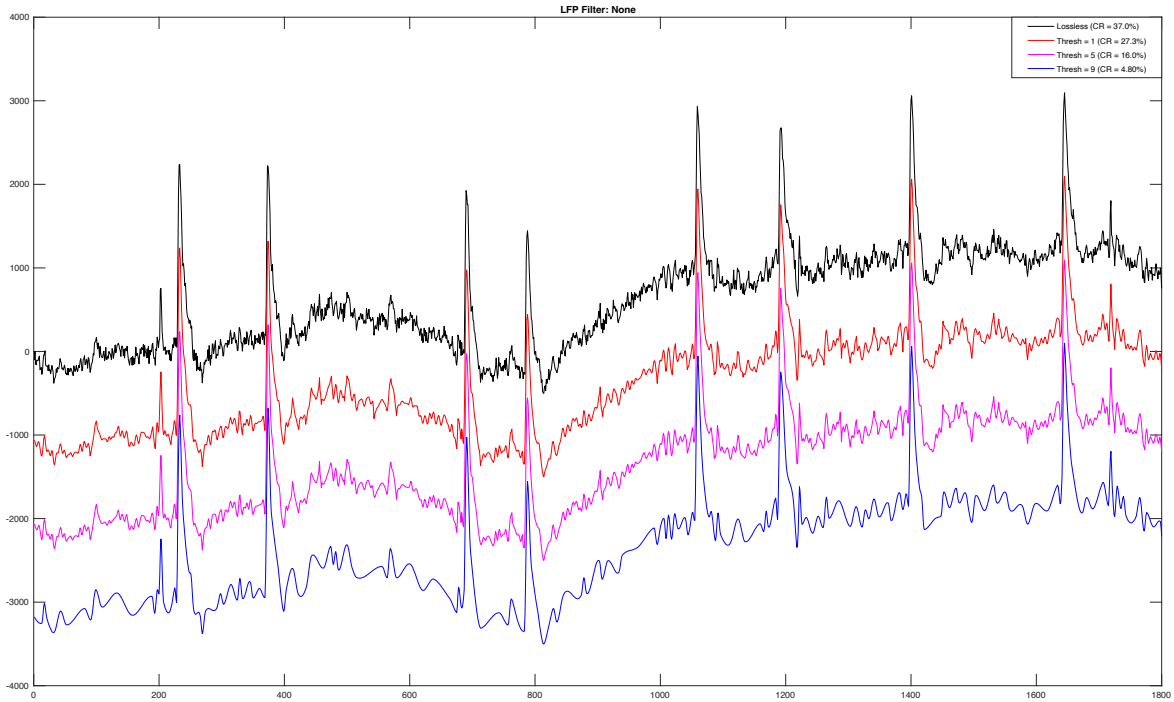
Typical VDS Lossy Waveform Results



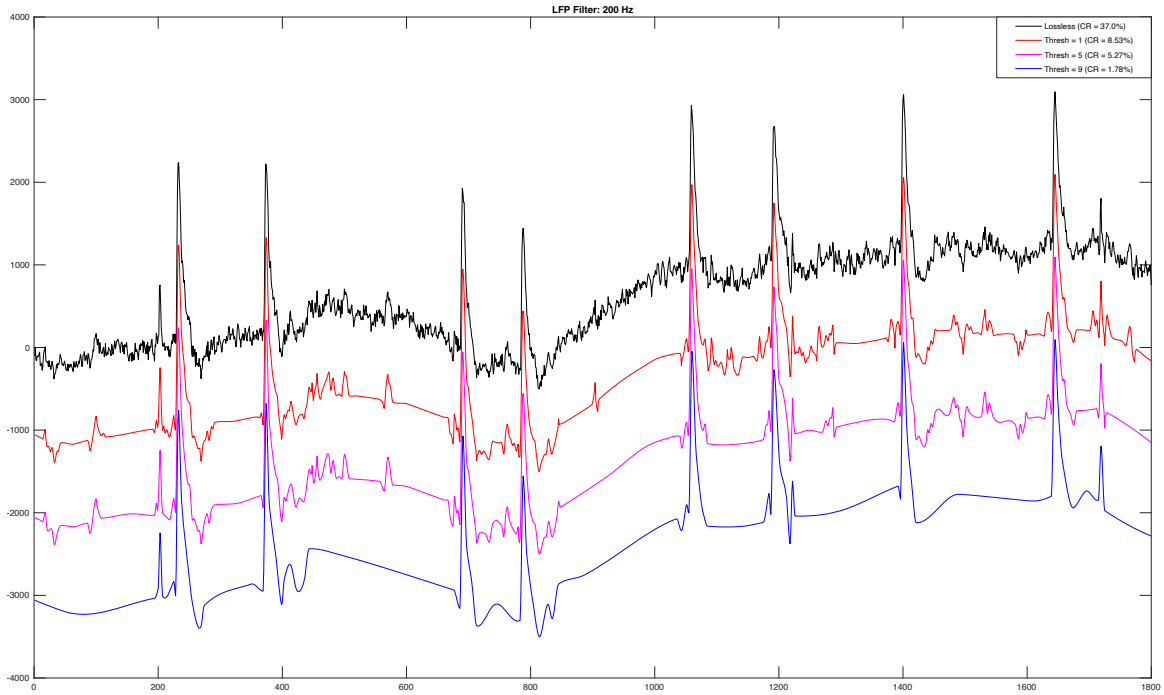
VDS Algorithm: Anchor Selection
(Compressed with thresh = 5, LFP high fc = 500 Hz)

The top traces are the raw data (black), & the VDS data (red) overlaid on it. The middle trace (black line) is the raw data with the sample points displayed (black dots). The bottom trace is the interpolated VDS data (red line) with the anchor points displayed (blue dots).

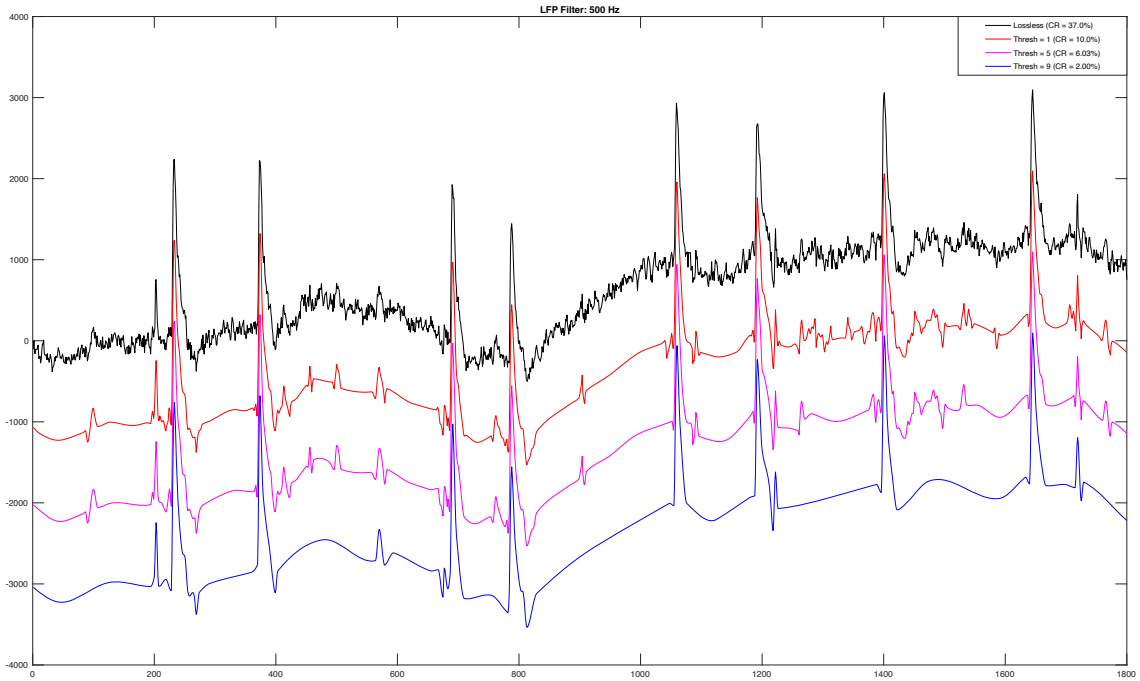
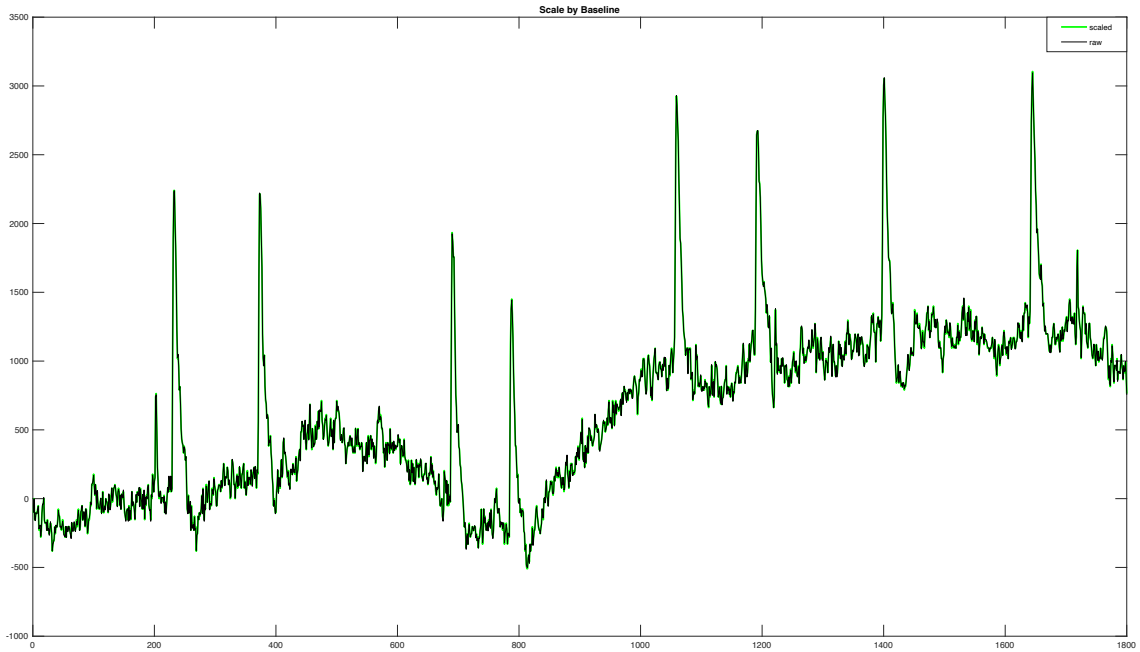
VDS Algorithm: No LFP Filter *(thresholds = 1, 5, & 9)*



VDS Algorithm: LFP Filter @ 200 Hz *(thresholds = 1, 5, & 9)*



VDS Algorithm: LFP Filter @ 500 Hz *(thresholds = 1, 5, & 9)*



VDS Algorithm: Scale by Baseline *(raw data)*